Sets

- **Definition 1**
  A **set** is a collection of objects, called **elements** of the set. If x is an element in a set A we denote it $x \in A$, and $x \notin A$ otherwise.
- **Example 1.1**
  If we have 3 pens in a pencil case x_1, x_2, x_3, then

$$B = \{x_1, x_2, x_3\} .$$

  And if we consider the blue pens in a pencil case

$$C = \{x \in B : x \text{ is blue}\} .$$

  For example, if x_1 and x_3 are blue pens we have that

$$C = \{x_1, x_3\} .$$

- **Definition 2**
  Let A be a set. We say that a set B is a **subset** of A, denoted $B \subseteq A$, if all the elements in B are also elements in A, that is, if $x \in B$ then $x \in A$ for all $x \in B$.

- **Definition 3**
  Let A and B be sets. We say that $A = B$, if $A \subseteq B$ and $B \subseteq A$.

- **Definition 4**
  If $B \subseteq A$ and $A \neq B$, we say that B is a **proper subset** of A and is denoted $B \subsetneq A$ (or $B \subset A$).
- **Example 4.1**
  Which of the following is a set?

$$\begin{array}{ll} (a)\ \{6,7,8,9,10\} & (b)\ \ \{\{e,d,f\},g,h\} \\ (c)\ \{e,d,f,g,h\} & (d)\ \{6,7,\{8,9\},10\} \end{array}$$

  a, b, c, d are sets.
- **Example 4.2**
  Note that

$$A = \{6,7,\{8,9\},10\}$$

  Determine if the following expressions are valid.

$$\begin{array}{ll} 6 \in A\ \checkmark & \{7,\{8,9\}\} \subseteq A\ \checkmark \\ 8 \in A\ \ \times & \{8,10\} \subseteq A\ \times \\ 8 \in A\ \ \times & \{\{8,9\},10\} \in A\ \times \\ \{8,9\} \in A\ \checkmark & \end{array}$$

- **Definition 5**
  The **empty set**, denoted $\varnothing$, is the set that has no elements.
- **Remark 5**
  Let A be a set, then $\varnothing \subseteq A$ and $A \subseteq A$.

- **Definition 6**
  Let A and B be two sets.
  The **union** of A and B, denoted $A \cup B$, is the set that contains all elements of A and all elements in B, that is,
  $$A \cup B = \{x : x \in A \text{ or } x \in B\}\,.$$
  The **intersection** of A and B, denoted $A \cap B$, is the set that contains all elements of A that are also elements of B, that is,
  $$A \cap B = \{x : x \in A \text{ and } x \in B\}\,.$$
  The **difference** A minus B, denoted $A \setminus B$, is the set that contains all elements of A that are not in B, that is,
  $$A \setminus B = \{x : x \in A \text{ and } x \notin B\}\,.$$
- **Example 6.1**
  Consider the sets $A = \{a, b, c, d, e\}$ and $B = \{b, c\}$, then
  $$\begin{aligned} A \cup B &= \{a, b, c, d, e\} \\ A \cap B &= \{b, c\} \\ A \setminus B &= \{a, d, e\} \\ B \setminus A &= \varnothing = \{\ \}\,. \end{aligned}$$

- **Definition 7**
  Let X be a set and consider two subsets A_1 and A_2 of the set X.
  We say that A_1 and A_2 are **disjoint** if $A_1 \cap A_2 = \varnothing$.
  We say that A_1 and A_2 form a **partition** of X if $X = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$ with $A_1, A_2 \neq \varnothing$.

- **Definition 8**
  Let A be a set, the **complement** of A, denoted A^{C}, is the set $A^{\mathsf{C}} = \{x : x \notin A\}$.
  - **Proposition 8.1**
    Let X be a set.
    1. If $A, B \subseteq X$ then $A, B \subseteq A \cup B$ and $A \cap B \subseteq A, B$.
    2. If $A, B \subseteq X$ then $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
    3. If $A, B \subseteq X$ and $A \subseteq B$ then $B^{\mathsf{C}} \subseteq A^{\mathsf{C}}$.
    4. $X^{\mathsf{C}} = \varnothing$ and $\varnothing^{\mathsf{C}} = X$.
    5. If $A, B \subseteq X$ then $(A \cup B)^{\mathsf{C}} = A^{\mathsf{C}} \cap B^{\mathsf{C}}$.
    6. If $A, B \subseteq X$ then $(A \cap B)^{\mathsf{C}} = A^{\mathsf{C}} \cup B^{\mathsf{C}}$.

    $5, 6$ are De Morgan's laws.

    Proof
    - Let $x \in A$, then $x \in A$ or $x \in B$, that is,
      $$x \in A \cup B \implies A \subseteq A \cup B.$$
      Analogously, we have that $B \subseteq A \cup B$.
      Let $x \in A \cap B$, then $x \in A$ and $x \in B$, that is,
      $$x \in A \cap B \implies A \cap B \subseteq A, B.$$
    - $$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\} \\ &= \{x : x \in B \text{ or } x \in A\} = B \cup A \end{aligned}.$$

Analogously, we have that $A \cap B = B \cap A$.

- Suppose that $A \subseteq B$ and let $x \in B^{\mathsf{C}}$, then $x \notin B$; since $A \subseteq B$ we have that $x \notin A$ and thus $x \in A^{\mathsf{C}}$.
  Therefore, $B^{\mathsf{C}} \subseteq A^{\mathsf{C}}$.
- By definition:

$$\begin{aligned} X^{\mathsf{C}} &= \{x \in X : x \notin X\} = \varnothing \\ \varnothing^{\mathsf{C}} &= \{x \in X : x \notin \varnothing\} = X. \end{aligned}$$

- Let $x \in (A \cup B)^{\mathsf{C}}$, then

$$\begin{aligned} x \in (A \cup B)^{\mathsf{C}} &\iff x \notin (A \cup B) \\ &\iff \text{It is false that } x \in A \vee x \in B \\ &\iff \text{It is true that } x \notin A \wedge x \notin B \\ &\iff x \notin A \text{ and } x \notin B \\ &\iff x \in A^{\mathsf{C}} \text{ and } x \in B^{\mathsf{C}} \\ &\iff x \in A^{\mathsf{C}} \cap B^{\mathsf{C}}. \end{aligned}$$

Thus $(A \cup B)^{\mathsf{C}} = A^{\mathsf{C}} \cap B^{\mathsf{C}}$.

- Let $x \in (A \cap B)^{\mathsf{C}}$, then

$$\begin{aligned} x \in (A \cap B)^{\mathsf{C}} &\iff x \notin (A \cap B) \\ &\iff \text{It is false that } x \in A \wedge x \in B \\ &\iff \text{It is true that } x \notin A \vee x \notin B \\ &\iff x \notin A \text{ or } x \notin B \\ &\iff x \in A^{\mathsf{C}} \text{ or } x \in B^{\mathsf{C}} \\ &\iff x \in A^{\mathsf{C}} \cup B^{\mathsf{C}}. \end{aligned}$$

Thus $(A \cap B)^{\mathsf{C}} = A^{\mathsf{C}} \cup B^{\mathsf{C}}$.

- **Definition 9**
  Let X be a set. The **power set** of X, denoted $\mathscr{P}(X)$, is the set of all subsets of X.
  - **Example 9.1**
    Let $X = \{a, b, c\}$, then

$$\mathscr{P}(X) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

  - **Example 9.2**
    Let A and B be sets. Prove that $\mathscr{P}(A) \subseteq \mathscr{P}(B)$, if and only if, $A \subseteq B$.
    - **Proof 9.2.1**
      Suppose that $\mathscr{P}(A) \subseteq \mathscr{P}(B)$. Let $x \in A$ then $\{x\} \in \mathscr{P}(A)$ and thus $\{x\} \in \mathscr{P}(B)$, therefore $x \in B$, hence $A \subseteq B$.
      Suppose that $A \subseteq B$. Let $X \in \mathscr{P}(A)$ then $X \subseteq A$, that is, $X \subseteq B$, therefore $X \in \mathscr{P}(B)$ and thus $\mathscr{P}(A) \subseteq \mathscr{P}(B)$.